The DIII Winding Number and the Massless Spin(2,3) Sector

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Abstract

We study the topological invariant naturally associated with the corrected DIII reading of the Spin(2,3) spinor sector. Starting from the explicit gamma-matrix representation, we construct a chiral four-generator Clifford frame that anticommutes with the grading operator $\Sigma = 2J^{01}$ and defines a flattened chiral Hamiltonian $Q(X)$ on S^3. In the chiral basis, Q has off-diagonal block $q(X) \in SU(2)$, and the DIII/AIII winding integral reduces to the degree of the map $q : S^3 \to SU(2) \cong S^3$. With the orientation and sign conventions adopted here, the invariant is

$$W_3 = -1.$$

The overall sign is convention-dependent, but the nontriviality of the invariant is not. The massless Spin(2,3) sector is therefore best understood not as a gapped phase carrying the invariant by itself, but as the equatorial slice inside the same nontrivial Spin(2,3) family.

1. Introduction

The purpose of this paper is narrow. We do not attempt a complete topological completion of the Spin(2,3) framework. Instead we isolate one technical question: once the corrected DIII class assignment is accepted, what is the corresponding $d = 3$ winding number?

This question matters because the existing topological spine already fixes most of the algebra. The spinor representation, the grading operator Σ, and the two relevant anti-unitary operators are already explicit. What remained open was the invariant itself. The aim here is to compute it in a convention-fixed, line-by-line way and to state clearly what the result does and does not imply.

The outcome is simple. The natural chiral Spin(2,3) Hamiltonian built from the Clifford frame defines a map from S^3 into $SU(2)$, and that map has degree -1 in the conventions used below. Reversing orientation or replacing q by $q^\dagger$ flips the sign, but not the fact that the invariant is nonzero. This is the robust content of the computation.

2. Spin(2,3) Conventions and the Corrected Class Assignment

We work with the four-component spinor representation

$$\gamma^0 = i\sigma^2 \otimes \mathbf{1}_2, \qquad \gamma^1 = i\sigma^1 \otimes \mathbf{1}_2,$$

$$\gamma^2 = \sigma^3 \otimes \sigma^1, \qquad \gamma^3 = \sigma^3 \otimes \sigma^2, \qquad \gamma^4 = \sigma^3 \otimes \sigma^3,$$

with signature $(2, 3)$.

The grading operator is taken to be

$$\Sigma = \sigma^3 \otimes \mathbf{1}_2,$$

which agrees with the sector split up to the harmless overall sign convention relating Σ and $2J^{01}$.

The relevant anti-unitary symmetries are

$$C = (\gamma^1\gamma^3)K,$$

and

$$T_0 = -i(\gamma^0\gamma^3)K,$$

with

$$C^2 = +1, \qquad T_0^2 = -1.$$

The phase of T_0 matters here: the operator is defined only up to an overall phase, but the above choice is the one for which the chiral operator is recovered exactly from C and T_0. Writing the anti-unitaries as $C = M_C K$ and $T_0 = M_{T_0} K$, with

$$M_C = \gamma^1\gamma^3, \qquad M_{T_0} = -i\gamma^0\gamma^3,$$

their product is linear and equals

$$CT_0 = M_C M_{T_0}^*.$$

In the present gamma-matrix representation one finds

$$M_C = \sigma^2 \otimes \sigma^2,$$

and

$$M_{T_0} = -i\gamma^0\gamma^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix},$$

so

$$CT_0 = M_C M_{T_0}^* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \Sigma.$$

Thus the chiral operator is not imported from outside the Spin(2,3) algebra: it is generated internally by the corrected combination of charge conjugation and partial time reversal. Forgetting the anti-unitary symmetries yields the chiral unitary class AIII; retaining them refines the same Hamiltonian family to DIII.

3. The Chiral Spin(2,3) Hamiltonian

Define the following Hermitian matrices:

$$\Gamma_0 = -i\gamma^0 = \sigma^2 \otimes \mathbf{1}_2,$$

$$\Gamma_1 = \gamma^0\gamma^2 = -\sigma^1 \otimes \sigma^1, \qquad \Gamma_2 = \gamma^0\gamma^3 = -\sigma^1 \otimes \sigma^2, \qquad \Gamma_3 = \gamma^0\gamma^4 = -\sigma^1 \otimes \sigma^3.$$

These obey

$$\Gamma_a^\dagger = \Gamma_a, \qquad \Gamma_a^2 = \mathbf{1}_4, \qquad \{\Gamma_a, \Gamma_b\} = 2\delta_{ab}\mathbf{1}_4,$$

for $a, b = 0, 1, 2, 3$, and each anticommutes with Σ:

$$\{\Sigma, \Gamma_a\} = 0.$$

Hence the Spin(2,3) data contains a natural chiral four-generator Clifford frame.

Let

$$X = (X_0, X_1, X_2, X_3) \in S^3, \qquad X_0^2 + X_1^2 + X_2^2 + X_3^2 = 1,$$

and define

$$Q(X) = X_0\Gamma_0 + X_1\Gamma_1 + X_2\Gamma_2 + X_3\Gamma_3.$$

Because the Γ_a mutually anticommute and square to the identity,

$$Q(X)^2 = \mathbf{1}_4,$$

so Q is already flattened.

This is the correct place where the invariant is defined. The strictly massless sector is recovered by the equatorial slice $X_0 = 0$, but the winding number itself belongs to the full gapped chiral family on S^3.

4. Chiral Basis and the $SU(2)$ Block

Since Q anticommutes with Σ, it is already off-diagonal in the chiral basis of $\Sigma = \sigma^3 \otimes \mathbf{1}_2$. In that basis one has

$$Q(X) = \begin{pmatrix} 0 & q_0(X) \\ q_0(X)^\dagger & 0 \end{pmatrix},$$

with

$$q_0(X) = -iX_0\mathbf{1}_2 - (X_1\sigma^1 + X_2\sigma^2 + X_3\sigma^3).$$

This follows directly from the block forms

$$\Gamma_0 = \begin{pmatrix} 0 & -i\mathbf{1}_2 \\ i\mathbf{1}_2 & 0 \end{pmatrix}, \qquad \Gamma_j = \begin{pmatrix} 0 & -\sigma^j \\ -\sigma^j & 0 \end{pmatrix}, \quad j = 1, 2, 3,$$

so the upper-right block of $Q(X) = X_0\Gamma_0 + X_1\Gamma_1 + X_2\Gamma_2 + X_3\Gamma_3$ is exactly the stated $q_0(X)$.

Now introduce the constant unitary

$$U = \begin{pmatrix} e^{-i\pi/4}\mathbf{1}_2 & 0 \\ 0 & e^{i\pi/4}\mathbf{1}_2 \end{pmatrix}.$$

Then

$$U^\dagger Q(X) U = \begin{pmatrix} 0 & e^{i\pi/2} q_0(X) \\ e^{-i\pi/2} q_0(X)^\dagger & 0 \end{pmatrix},$$

and since $e^{i\pi/2} = i$, the upper-right block becomes

$$q(X) = iq_0(X) = X_0\mathbf{1}_2 - i(X_1\sigma^1 + X_2\sigma^2 + X_3\sigma^3).$$

Thus after this explicit constant chiral rotation, Q takes the standard form

$$Q(X) = \begin{pmatrix} 0 & q(X) \\ q(X)^\dagger & 0 \end{pmatrix},$$

with

$$q(X) = X_0 \mathbf{1}_2 - i(X_1\sigma^1 + X_2\sigma^2 + X_3\sigma^3).$$

Because $X \in S^3$,

$$q(X)^\dagger q(X) = \mathbf{1}_2, \qquad \det q(X) = 1,$$

so

$$q : S^3 \to SU(2) \cong S^3.$$

The winding number therefore reduces to the degree of this map. In the chiral formulation the invariant is

$$W_3 = \frac{1}{24\pi^2} \int_{S^3} \mathrm{tr}\Big[(q^{-1}dq)^3\Big].$$

5. Explicit Computation of W_3

Parameterize S^3 by

$$X_0 = \cos\chi, \qquad X_1 = \sin\chi\cos\theta,$$
$$X_2 = \sin\chi\sin\theta\cos\phi, \qquad X_3 = \sin\chi\sin\theta\sin\phi,$$

with

$$0 \le \chi \le \pi, \qquad 0 \le \theta \le \pi, \qquad 0 \le \phi < 2\pi.$$

Write

$$q = X_0\mathbf{1}_2 - i\vec{X}\cdot\vec{\sigma}, \qquad q^{-1} = q^\dagger = X_0\mathbf{1}_2 + i\vec{X}\cdot\vec{\sigma}.$$

Using the Pauli identity

$$\sigma_i\sigma_j = \delta_{ij}\mathbf{1}_2 + i\epsilon_{ijk}\sigma_k,$$

one finds

$$q^{-1}dq = (X_0\mathbf{1}_2 + iX_i\sigma_i)(dX_0\mathbf{1}_2 - i\,dX_j\sigma_j).$$

Expanding and using $X_0\,dX_0 + X_i\,dX_i = 0$ on S^3, the scalar part vanishes and the matrix part reduces to

$$q^{-1}dq = i\alpha_i\sigma_i,$$

where

$$\alpha_i = X_i\,dX_0 - X_0\,dX_i + \epsilon_{ijk}X_j\,dX_k.$$

Equivalently, defining $\omega_i = -\alpha_i$, one may write

$$q^{-1}dq = -i\,\omega_i\sigma_i,$$

where the one-forms ω_i are the standard left-invariant coframe on $SU(2) \cong S^3$.

The cubic trace is then obtained directly from

$$\mathrm{tr}(\sigma_i\sigma_j\sigma_k) = 2i\epsilon_{ijk},$$

which gives

$$\mathrm{tr}\Big[(q^{-1}dq)^3\Big] = 2\,\epsilon_{ijk}\,\alpha_i\wedge\alpha_j\wedge\alpha_k.$$

Substituting the coordinate parameterization of S^3, one obtains the explicit one-forms

$$\alpha_1 = -\cos\theta\,d\chi + \cos\chi\sin\chi\sin\theta\,d\theta + \sin^2\chi\sin^2\theta\,d\phi,$$

$$\alpha_2 = -\sin\theta\cos\phi\, d\chi + (-\cos\chi\sin\chi\cos\theta\cos\phi - \sin^2\chi\sin\phi)\, d\theta + (\cos\chi\sin\chi\sin\theta\sin\phi - \sin^2\chi\sin\theta\cos\theta\cos\phi)\, d\phi,$$

$$\alpha_3 = -\sin\theta\sin\phi\, d\chi + (-\cos\chi\sin\chi\cos\theta\sin\phi + \sin^2\chi\cos\phi)\, d\theta + (-\cos\chi\sin\chi\sin\theta\cos\phi - \sin^2\chi\sin\theta\cos\theta\sin\phi)\, d\phi.$$

Their coefficient matrix in the basis $(d\chi, d\theta, d\phi)$ has determinant

$$\det({\alpha_i}^\mu) = -\sin^2\chi\sin\theta,$$

and hence

$$\epsilon_{ijk}\,\alpha_i \wedge \alpha_j \wedge \alpha_k = -6\sin^2\chi\sin\theta\, d\chi \wedge d\theta \wedge d\phi.$$

Therefore

$$\mathrm{tr}\Big[(q^{-1}dq)^3\Big] = -12\sin^2\chi\sin\theta\, d\chi \wedge d\theta \wedge d\phi.$$

With the domain orientation taken to be $d\chi \wedge d\theta \wedge d\phi$, this is the sign entering the winding integral in the present convention. This coordinate orientation is the standard outward orientation on S^3: indeed

$$\det\big(X, \partial_\chi X, \partial_\theta X, \partial_\phi X\big) = \sin^2\chi\sin\theta > 0$$

away from the coordinate singular sets. The minus sign in W_3 therefore does not come from the parameterization of S^3; it comes from the winding-number convention used here, namely

$$W_3 = \frac{1}{24\pi^2}\int_{S^3} \mathrm{tr}\Big[(q^{-1}dq)^3\Big].$$

Many references instead insert the opposite overall sign, or equivalently use the opposite Maurer-Cartan convention, in which case the same map $q(X) = X_0\mathbf{1}_2 - i\vec{X}\cdot\vec{\sigma}$ would be assigned $+1$.

Therefore

$$W_3 = \frac{1}{24\pi^2}\int_0^\pi\int_0^\pi\int_0^{2\pi} (-12)\sin^2\chi\sin\theta\, d\phi\, d\theta\, d\chi.$$

Evaluating the integrals,

$$\int_0^\pi \sin^2\chi\, d\chi = \frac{\pi}{2}, \qquad \int_0^\pi \sin\theta\, d\theta = 2, \qquad \int_0^{2\pi} d\phi = 2\pi,$$

so

$$W_3 = \frac{-12}{24\pi^2}\left(\frac{\pi}{2}\right)(2)(2\pi) = -1.$$

Thus the topological invariant is nontrivial. Its overall sign depends on orientation and on the sign convention in the definition of q; replacing q by $q^\dagger$, or reversing orientation on S^3, yields $+1$ instead. The convention-independent statement is therefore:

the Spin(2,3) chiral Hamiltonian carries $|W_3| = 1$.

6. Interpretation of the Massless Sector

The computation above applies to the gapped chiral family $Q(X)$ on S^3. The strictly massless sector corresponds to the equatorial slice

$$X_0 = 0,$$

for which

$$q = -i(X_1\sigma^1 + X_2\sigma^2 + X_3\sigma^3).$$

This equator is not itself a gapped three-dimensional DIII phase carrying W_3. The computation above therefore does not by itself prove a full interface theorem. What it does show is that the massless slice sits naturally as the equator of a gapped chiral family with nontrivial unit winding. It is therefore reasonable to regard it as a candidate transition locus within that family, but that last interpretive step goes beyond the bare invariant calculation.

The relation to the earlier AIII language is then straightforward. If one suppresses the anti-unitary symmetries and keeps only the chiral grading, the same flattened Hamiltonian lies in class AIII. Including T_0 and C refines that same chiral Hamiltonian to class DIII. The integer computed here is therefore the same winding number under both descriptions, even though AIII and DIII classify different symmetry-constrained Hamiltonian families. It is most precise to present the result in DIII language because the corrected Spin(2,3) symmetry data contains those anti-unitaries explicitly.

7. What the Result Does and Does Not Establish

The result established here is exact and narrow. The corrected Spin(2,3) symmetry data yields a natural chiral Clifford frame, that frame defines a flattened Hamiltonian on S^3, and the associated winding number is nontrivial, equaling -1 in the present orientation convention. This is the firm mathematical content of the paper.

What is not established is equally important. The calculation does not by itself provide a material realization of the corresponding DIII Hamiltonian, an anomaly-inflow statement, a generation-counting theorem from topology, or a full extension of the analysis to the broader two-time-reversal picture. Those questions remain open and should be treated separately from the invariant computed here.

8. Numerical Cross-Check

A numerical integration of the winding formula using a direct discretization of the S^3 coordinates reproduces the analytic result. With the current script and a uniform $28 \times 28 \times 28$ grid in (χ, θ, ϕ), one finds

$$W_3 \approx -0.999.$$

The residual error is discretization error and decreases under grid refinement. A script implementing this check is included with the source files for this paper.

9. Conclusion

The corrected topological Spin(2,3) story now has a concrete invariant. The natural chiral Hamiltonian extracted from the explicit gamma-matrix representation defines a map $S^3 \to SU(2)$ of degree one in magnitude, giving $W_3 = -1$ in the conventions adopted here. The strictly massless sector is therefore best viewed as the equatorial slice of a gapped chiral family with nontrivial unit winding, rather than as an isolated gapped phase carrying the invariant on its own. Interpreting that slice as a genuine transition locus is plausible, but it remains one step beyond what the winding-number calculation alone proves.